

Lesson 03: Checking Solutions in Differential Equations (Explicit, Implicit & Families)

Professor Leonard Differential Equations Master Series · Lesson 3

- **Direct Video Link:** [Lesson 03: Checking Solutions in Differential Equations \(Explicit, Implicit & Families\)](#) · Duration: 30:58
- **Target Exam Scope:** 🎯 MIDTERM EXAM (Ch 1.2 — Solutions & Verification)
- **Course Context:** Applied Ordinary Differential Equations (ENGR 213)

1. Whiteboard Lecture Intuition & Philosophical Framing

In this 31-minute whiteboard lecture, Professor Leonard teaches how to rigorously verify whether a candidate function satisfies a given differential equation. He demonstrates the structural difference between explicit solutions, implicit relations, and multi-parameter general families, emphasizing that verifying solutions is the ultimate sanity check for every exam problem.

2. Core Theoretical Concepts & Mathematical Formulations

1. Explicit Solutions vs. Implicit Solutions

- **Explicit Solution:** A function where the dependent variable is isolated entirely on one side of the equality:

$$y = \phi(x) \quad \text{for all } x \in I$$

- **Implicit Solution:** A relation $G(x, y) = 0$ that defines one or more explicit solutions implicitly on an interval I , requiring implicit differentiation to verify:

$$\frac{d}{dx}[G(x, y)] = 0 \implies \text{produces the original ODE}$$

2. General Solutions vs. Particular Solutions

- **General Solution:** An n -parameter family of functions $y = \phi(x, c_1, c_2, \dots, c_n)$ containing n independent arbitrary constants, where n is equal to the order of the ODE.
 - **Particular Solution:** A specific trajectory obtained by assigning explicit numerical values to the arbitrary constants $c_1, \dots, c_n$, determined by initial or boundary constraints.
 - **Singular Solution:** A valid solution that **cannot** be obtained from the general family for any finite choice of parameters c_i (frequently arising as envelope curves or lost division asymptotes).
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3. Classroom Chalkboard Examples (Solved in Complete Baby Steps)

Problem 03.1: Chalkboard Problem 3.1: Explicit Verification of a Two-Parameter Family

Problem Statement:

Verify that the two-parameter family $y(x) = c_1 e^{2x} + c_2 x e^{2x}$ is an explicit solution to the homogeneous 2nd-order ODE: $y'' - 4y' + 4y = 0$ for all real constants c_1, c_2 .

Step-by-Step Whiteboard Solution:

- **Step 1: Compute the First Derivative $y'(x)$ using the Product Rule:**

$$y(x) = c_1 e^{2x} + c_2 x e^{2x}$$

$$y'(x) = 2c_1 e^{2x} + c_2 [1 \cdot e^{2x} + x(2e^{2x})] = (2c_1 + c_2)e^{2x} + 2c_2 x e^{2x}$$

- **Step 2: Compute the Second Derivative $y''(x)$:**

$$y''(x) = 2(2c_1 + c_2)e^{2x} + 2c_2 [1 \cdot e^{2x} + x(2e^{2x})]$$

$$y''(x) = (4c_1 + 2c_2)e^{2x} + 2c_2 e^{2x} + 4c_2 x e^{2x} = (4c_1 + 4c_2)e^{2x} + 4c_2 x e^{2x}$$

- **Step 3: Substitute y, y', y'' into the Left-Hand Side (LHS) of the ODE:**

$$\text{LHS} = y'' - 4y' + 4y$$

$$\text{LHS} = [(4c_1 + 4c_2)e^{2x} + 4c_2 x e^{2x}] - 4[(2c_1 + c_2)e^{2x} + 2c_2 x e^{2x}] + 4[c_1 e^{2x} + c_2 x e^{2x}]$$

- **Step 4: Group Terms by Basis Functions (e^{2x} and $x e^{2x}$):**

- Aggregate e^{2x} coefficients:

$$(4c_1 + 4c_2) - 4(2c_1 + c_2) + 4c_1 = 4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1 = (4 - 8 + 4)c_1 + (4 - 4)c_2 = 0$$

- Aggregate $x e^{2x}$ coefficients:

$$4c_2 - 4(2c_2) + 4c_2 = 4c_2 - 8c_2 + 4c_2 = (4 - 8 + 4)c_2 = 0$$

- **Step 5: Verify Equality:**

$$\text{LHS} = 0 \cdot e^{2x} + 0 \cdot x e^{2x} = 0 = \text{RHS} \quad \checkmark$$

Therefore, $y(x) = c_1 e^{2x} + c_2 x e^{2x}$ is an explicit solution on $(-\infty, \infty)$ for any $c_1, c_2 \in \mathbb{R}$.

[!WARNING] **Common Exam Pitfall:** Failing to apply the product rule to terms like $c_2 x e^{2x}$. Differentiating $x e^{2x}$ as just e^{2x} will completely derail the cancellation.

Problem 03.2: Chalkboard Problem 3.2: Implicit Verification and Domain Analysis

Problem Statement:

Verify that the circle relation $x^2 + y^2 = 25$ defines an implicit solution to the first-order ODE $\frac{dy}{dx} = -\frac{x}{y}$. Determine the valid interval of definition.

Step-by-Step Whiteboard Solution:

- **Step 1: Differentiate the Relation Implicitly with Respect to x :**

$$\begin{aligned}\frac{d}{dx} [x^2 + y^2] &= \frac{d}{dx} [25] \\ 2x + 2y \frac{dy}{dx} &= 0\end{aligned}$$

- **Step 2: Solve Algebraically for dy/dx :**

$$2y \frac{dy}{dx} = -2x \implies \frac{dy}{dx} = -\frac{x}{y}$$

This identically reproduces the original differential equation!

- **Step 3: Analyze the Domain and Singularities:** Solving $x^2 + y^2 = 25$ explicitly for y yields two separate branches:

$$y_1(x) = +\sqrt{25 - x^2}, \quad y_2(x) = -\sqrt{25 - x^2}$$

- For either branch, the derivative is $\frac{dy}{dx} = \mp \frac{x}{\sqrt{25 - x^2}} = -\frac{x}{y}$.
- At $x = \pm 5, y = 0$, causing the denominator in $dy/dx = -x/y$ to vanish (vertical tangent lines).
- Therefore, the solution is valid on the open interval $I = (-5, 5)$.

[!WARNING] **Common Exam Pitfall:** Claiming the solution is valid on the closed interval $[-5, 5]$. Solutions to differential equations must be differentiable throughout their interval of definition; infinite derivatives at boundaries violate this condition.

4. Key Takeaways & Exam Strategy Formula Card

- Watch the full whiteboard demonstration on YouTube: [Lesson 03: Checking Solutions in Differential Equations \(Explicit, Implicit & Families\)](#).
- Master the classification and verification mechanics before attempting complex integrations.
- Always check domain restrictions and verify initial values back into the computed trajectory.